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## Bulk material receiving, conveying, storing, and dispensing

### Abstract

A bulk material handling system includes a majors material handling system including bulk material storage modules and bulk material dispensing modules. The dispensing modules include bulk material dosing assemblies and docking assemblies. A bulk material handling method includes conveying bulk material from a mobile bulk material container into a stationary bulk material container at a glass manufacturing facility via dense phase pneumatic conveying.

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## Background/Summary

TECHNICAL FIELD

(1) This patent application discloses innovations to material handling and, more particularly, to bulk material handling including receiving, conveying, storing, and dispensing of bulk materials.

## BACKGROUND

(2) A conventional glass "batch house" includes a custom architectural installation specifically designed for glass manufacturing, and a glass batch handling system supported and sheltered by the architectural installation. The batch house is generally configured to receive and store glass feedstock, or "glass batch" materials, including glassmaking raw materials, for example, sand, soda ash, and limestone, and also including cullet in the form of recycled, scrap, or waste glass. The conventional glass batch house requires a specialized, dedicated, and permanent architectural installation including a tall building and a covered receiving platform and pit to receive glass batch from underneath railcars or trucks that arrive loaded with glass batch materials. The batch house also includes multi-story silos to store the glass batch, and glass batch elevators and conveyors to move the glass batch from receiving systems at a bottom of the pit to tops of the silos. The batch house further includes cullet pads at ground level to receive and store cullet, crushers to crush cullet to a size suitable for melting, and cullet elevators and conveyors to move crushed cullet to one of the silos in the batch house. The batch house additionally includes a mixer to mix the glass batch received from the silos, conveyors integrated with scales to weigh and deliver each glass batch material from the silos to the mixer, mixer conveyors to move the glass batch from the mixers to the hot-end subsystem, and dust collectors to collect dust from the various equipment. The installation occupies a large footprint and a large volumetric envelope, takes about one to two years to construct, cannot be relocated from one location to another, and tends to be a dusty and dirty environment.

## SUMMARY OF THE DISCLOSURE

(3) The present disclosure embodies a number of aspects that can be implemented separately from or in combination with each other.

(4) Embodiments of a bulk material storage module include a container module frame, a bulk material container supported within the frame, the bulk material container having: an upper portion and a lower portion, an inlet located along the upper portion for receiving bulk material into the material container, an outlet located along the lower portion for discharging bulk material from the material container, and a vent to permit air exchange between an inside of the container and outside the container during receiving and/or discharging of bulk material from the material container. The module further includes at least one utilities receiver configured to couple the module with at least one of: a control system, an electric utility, a pneumatic utility, or another bulk material storage module. The module is configured to be attached side-by-side with up to four other bulk material storage modules and corner-to-corner with up to four other bulk material storage modules, all of the modules having identical frames and bulk material containers.

(5) Embodiments of a bulk material dispensing module include: a dispensing module frame having a longitudinal axis, the frame further comprising a plurality of transverse frame members spaced along the longitudinal axis, wherein a dispensing cell is defined between each pair of transverse frame members; at least one bulk material dispenser supported within the frame, each bulk material dispenser being supported in a different dispensing cell and comprising: an inlet accessible through a first side of the frame and configured to be coupled with and receive material from a bulk material container, an outlet accessible through an opposite side of the frame and configured to be coupled with and discharge material to a transport bin, and a conveyor configured to move bulk material from the inlet to the outlet when the inlet is coupled with the bulk material container. The module further includes a controller carried by the frame for each bulk material dispenser. The module is configured to be attached side-by-side with one or more other bulk material dispensing modules, each of the modules having identical frames, dispenser inlets, and dispenser outlets, and the storage module has external dimensions less than or equal to an intermodal freight container.

(6) Embodiments of a bulk material handling method include conveying bulk material directly from a mobile bulk material container into a stationary bulk material container at a glass manufacturing facility via dense phase pneumatic conveying.

(7) Embodiments of a bulk material dispenser include a dispenser inlet configured for coupling with and receiving bulk material from an outlet of a bulk material container, a dispenser outlet configured for coupling with and discharging the bulk material into a transport bin, a conveyor that moves bulk material received at the inlet side toward the outlet, and a filter assembly configured to filter solids from air displaced from the transport bin during dispenser operation.
(8) Embodiments of a docking assembly for use in a bulk material dispensing system include an inlet configured for coupling with and receiving bulk material from a bulk material dosing assembly, and an outlet configured for coupling with and discharging the bulk material into a transport bin. The outlet is moveable toward and away from the inlet and, thereby, respectively away from and toward the transport bin.
(9) Embodiments of a bulk material handling method include: coupling an outlet of a bulk material dispenser with a transport bin to form a closure at an inlet of the transport bin and place an inside of the transport bin in communication with the dispenser; receiving bulk material in the dispenser from a bulk material container; forming a reduced pressure region in an internal volume of the dispenser; and dispensing the bulk material from the dispenser and into the transport bin through the reduced pressure region.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a perspective view of a bulk material handling system in accordance with an illustrative embodiment of the present disclosure, illustrating a building having a roof, cladding, elevator, stairs, ladders, and platforms.
(2) FIG. **1**B is another perspective view of the system corresponding to FIG. **1**A, without the roof, cladding, elevator, and ladders.
(3) FIG. **2**A is a different perspective view of the system of FIG. **1**A, illustrating the building with the roof, cladding, elevator, stairs, ladders, and platforms.
(4) FIG. **2**B is another perspective view of the system corresponding to FIG. **2**A, without the roof, cladding, elevator, and ladders.
(5) FIG. **3** is a top view of the system of FIG. **1**A.
(6) FIG. **4** is a bottom view of the system of FIG. **1**A.
(7) FIG. **5** is an end view of the system of FIG. **1**A.
(8) FIG. **6** is another end view of the system of FIG. **1**A opposite that of FIG. **5**.
(9) FIG. **7** is a side view of the system of FIG. **1**A.
(10) FIG. **8** is another side view of the system of FIG. **1**A opposite that of FIG. **7**.
(11) FIG. **9** is a perspective view of a modular frame of the system of FIG. **1**A.
(12) FIG. **10** is a perspective view of another modular frame of the system of FIG. **1**A.
(13) FIG. **11** is a perspective view of another modular frame of the system of FIG. **1**A.
(14) FIG. **12** is a perspective view of a bulk material storage module as part of a rack and module assembly.
(15) FIG. **13** is a perspective view of an illustrative modular bulk material storage and dispensing system.
(16) FIG. **14** is a partially exploded isometric view of the system of FIG. **13**.
(17) FIG. **15** is a further exploded isometric view of the system of FIG. **13**.
(18) FIG. **16** is an isometric view of an array of another modular bulk material storage and dispensing system.
(19) FIG. **17** is a perspective view of a bulk material storage module of the system of FIG. **13** in a shipping orientation as viewed from a top end of the module.
(20) FIG. **18** is a perspective elevation view of a bulk material storage container of the module of

FIG. **17**.

(21) FIG. **19** is a top perspective view of an array of storage container modules of the system of FIG. **13** with a top portion of the module frames omitted.

(22) FIG. **20**A is the installation of FIG. **1**A illustrated with a mobile bulk material container arrived at the installation to convey bulk material to the stationary bulk material containers of the installation with elements of a pneumatic conveying system additionally illustrated.

(23) FIG. **20**B is an elevation view of the inlet conduits of FIG. **20**A, further illustrating high-pressure lines and pulse-pressure lines for use in dense phase conveying of bulk material.

(24) FIG. **20**C is an isometric view of a pneumatic panel configured to provide elements of the dense phase conveying system.

(25) FIG. **20**D is a schematic representation of a mobile bulk material storage container coupled with the pneumatic dense phase conveying system.

(26) FIG. **21** is a schematic depiction of a conduit system with incoming bulk material routed to one of three storage containers of one branch of the conduit system.

(27) FIG. **22** is a schematic depiction of the conduit system of FIG. **21** with the incoming bulk material re-routed to a different one of the three storage containers.

(28) FIG. **23** is a schematic depiction of the conduit system of FIG. **21** with the incoming bulk material re-routed to a third one of the three storage containers.

(29) FIG. **24** is a top perspective view of a bulk material dispensing module of the system of FIG. **13**.

(30) FIG. **25** is a bottom perspective view of the bulk material dispensing module of the system of FIG. **13**.

(31) FIG. **26** is a perspective view of a bulk material dispenser of the module of FIGS. **24** and **25**.

(32) FIG. **27** is another perspective view of the bulk material dispenser of FIG. **26**.

(33) FIG. **28** is a perspective view of the bulk material dispenser of FIGS. **24-27** coupled with a transport bin.

(34) FIG. **29** is a top perspective view of the transport bin of FIG. **28** uncoupled from the bulk material dispenser.

(35) FIG. **30** is a perspective view of a docking assembly of the bulk material dispenser of FIG. **28** coupled with the transport bin.

(36) FIG. **31** is a schematic cross-sectional view of a screw conveyor of the bulk material dispenser of FIGS. **26-28**.

(37) FIG. **32** is an isometric view of a docking assembly.

(38) FIG. **33** is an isometric cross-sectional view of a docking assembly of the bulk material dispenser of FIGS. **26-28**.

(39) FIG. **34** is a schematic cross-sectional view of a portion of a docking assembly in a retracted condition over a transport bin.

(40) FIG. **35** is a schematic cross-sectional view of a portion of a docking assembly in an extended condition and coupled with a transport bin.

DETAILED DESCRIPTION

(41) In general, a new bulk material handling system is illustrated and described with reference to a glass feedstock handling system for a glass container factory as an example. Those of ordinary skill in the art would recognize that other glass factories, for example, for producing glass fibers, glass display screens, architectural glass, vehicle glass, or any other glass products, share many aspects with a glass container factory. Accordingly, the presently disclosed and claimed subject matter is not necessarily limited to glass containers, glass container feedstock handling systems, and glass container factories and, instead, encompasses any glass products, glass product feedstock handling systems, and glass product factories. Moreover, the presently disclosed and claimed subject matter is not necessarily limited to bulk material handling for the glass industry and, instead, encompasses any products, bulk material handling systems, and factories in any industry in which bulk material

handling is useful.
(42) Although conventional glass batch houses and methods enable efficient production of high-quality products for large-scale production runs, the presently disclosed subject matter facilitates implementation of a revolutionary bulk material handling system that is simpler than a conventional batch house, is modular and mobile, and is more compact and economical at least for smaller scale production runs or incremental additions to existing large-scale production runs. More specifically, in accordance with an aspect of the present disclosure, a new bulk material handling system may include prefabricated modular equipment configurations to facilitate rapid and mobile production capacity expansion in smaller increments and at lower capital cost than conventional glass batch houses, and also may include techniques for handling bulk material in a dust-free or reduced dust manner. Further, the new system may omit one or more conventional glass batch house subsystems or aspects thereof, as described in further detail below.
(43) With specific reference now to FIGS. **1**A through **8**, a new bulk material handling system **10** includes a new architectural installation **12** and new subsystems and equipment supported and sheltered by the installation **12**. The installation **12** includes a concrete foundation **14** having a floor which may include, for example, a four to six-inch-thick slab, and a bulk material handling building **16** on the foundation including walls **18** and a roof **20**. The installation **12** requires no basement and no pit below the floor, such that the concrete foundation has earthen material directly underneath, wherein the foundation slab establishes the floor. As used herein, the term "pit" includes an elevator pit, conveyor pit, loading pit, and the like, located below grade or below ground level and that may require excavation of earthen material. As used herein, the term "basement" includes the lowest habitable level of the bulk material handling building below a floor of the building and can include a first level or a below grade or below ground level portion that may require excavation of earthen material.
(44) The installation **12** also includes multiple habitable levels, including a base or first level **21**, an intermediate or second level **22**, an upper or third level **23**, and an attic or fourth level **24**. Also, as used herein, the term "habitable" means that there is standing room for an adult human in the particular space involved and there is some means of ingress/egress to/from the space while walking such as a doorway, stairway, and/or the like. The installation **12** further includes egress doors **26**, egress platforms **27**, stairs **28**, ladders **30**, and an elevator **32** to facilitate access to the egress platforms **27** and doors **26**. The installation **12** additionally includes loading doors **34**, loading platforms, and one or more ramps. Notably, the building **16** is constructed of many modules, including modular walls used to construct a base frame for the first level, and modular frames for the second, third, and fourth levels, as will be discussed in detail below.
(45) With continued reference to FIGS. **1**A through **8**, the bulk material handling system **10** includes several subsystems that occupy a volumetric envelope much smaller than conventional batch houses such that the system **10** likewise requires a smaller volumetric envelope than conventional glass batch houses. The bulk material handling system **10** may be a glass bulk material handling system configured to receive and store glass feedstock or "glass batch." The glass batch includes glassmaking raw materials, including glass feedstock "majors" and "minors" and also may include cullet in the form of recycled, scrap, or waste glass. The bulk material handling system receives glass batch bulk materials and combines them into doses and provides the doses to a downstream hot-end system of a glass factory adjacent to or part of the bulk material handling system.
(46) The bulk material handling system **10** includes one or more of the following subsystems. A first bulk material, or majors, subsystem **38** is configured to receive, pneumatically convey, store, and gravity dispense majors bulk material. Glassmaking majors may include sand, soda, limestone, alumina, saltcake, and, in some cases, dust recovery material. Similarly, a second bulk material, or minors, subsystem **40** is configured to receive, pneumatically convey, and store minors bulk material from individual bulk material bags. Glassmaking minors may include selenium, cobalt

oxide, and any other colorants, decolorants, fining agents, and/or other minors materials suitable for glassmaking. A bulk material discharging subsystem **54** is configured to receive bulk material from the majors and minors subsystems **38**, **40** and transmit the bulk material to downstream bulk material processing equipment, for example, a glass melting furnace separate from and downstream of the bulk material handling system **10**. A bulk material transfer or transport subsystem **44** is configured to receive bulk material from the majors and minors subsystems **38**, **40**, and transport the bulk material within, to, and from, the majors and minors subsystems **38**, **40**, and to and from the discharge subsystem **42**. A controls subsystem **46** is in communication with various equipment of one or more of the other subsystems **38**, **40**, **42**, **44**, and is configured to control various aspects of the system **10**. Those of ordinary skill in the art would recognize that the system **10** can be supplied with utility or plant electrical power, and can include computers, sensors, actuators, electrical wiring, and the like to power and communicate different parts of the system **10** together. Likewise, the system **10** can be supplied with plant or compressor pneumatic power/pressure, and can include valves, lubricators, regulators, conduit, and other like pneumatic components to pressurize and communicate different parts of the system **10** together.

(47) The system **10** may be pneumatically closed from pneumatic input or receiving conduit **39** of the majors subsystem **38** to pneumatic output or transmitting conduit **43** of the discharging subsystem **54**. The pneumatic receiving conduit **39** may extend through one or more walls of the building for accessibility to bulk transporters, e.g., trucks or rail cars, that bring bulk materials and that may have pressurized vessels to assist with pneumatic receiving and conveying of bulk material. The receiving conduit **39** has any suitable couplings for coupling to bulk transporters in a pneumatically sealed manner, wherein the bulk transporters may have pumps, valves, and/or other equipment suitable to pressurize the receiving conduit to push bulk material into the majors subsystem **38** and/or the batch handling system **10** itself may include pumps, valves, pressurized plant air plumbing, and/or other equipment suitable to apply positive and/or negative (vacuum) pressure to the input conduit to push and/or pull bulk material into the majors and minors subsystems **38**, **40**.

(48) The transmitting conduit **43** may extend through one or more walls or the roof of the building for transmission to downstream bulk material processing equipment, for instance, in a hot end subsystem of a glass manufacturing system (not shown). For example, the transmitting conduit **43** is pneumatically sealingly coupled to a receiver hopper at a glass melter in the hot end subsystem. The conduit **43** may have any suitable couplings for coupling to the receiver hopper in a pneumatically sealed manner. Those of ordinary skill in the art would recognize that the bulk material handling system is pneumatically closed between the pneumatic receiving conduit and the pneumatic transmitting conduit. This is in contrast to conventional systems where bulk material is open to the surrounding environment. The phrase "pneumatically closed" means that the path, and the bulk materials following that path, from receiving conduit to transmitting conduit is/are enclosed, and not openly exposed to the surrounding environment, although not necessarily always sealed air-tight.

(49) With reference to FIG. **9**, a representative modular wall **48** of the first level **21** of the building is constructed as a rectangular truss, having a longitudinal axis L and a vertical axis V, and including lower and upper beams **48***a*,*b* extending longitudinally and being vertically opposed from one another. The wall **48** also includes vertically extending end posts **48***c* and intermediate posts **48***d* longitudinally between the end posts **48***c*, and struts **48***e* extending obliquely between the beams and connected to the posts **48***c*,*d*. The modular wall **48** is preassembled at an equipment fabricator, shipped from the fabricator to a product manufacturer, and is erected at the product manufacturer. The modular wall has exterior dimensions less than or equal to exterior dimensions of an intermodal freight container, more specifically, a height less than or equal to 9′ 6″ (2.896 m), a width less than or equal to 8′ 6″ (2.591 m), and a length less than or equal to 53′ (16.154 m). As best illustrated in FIGS. **2**B, **7** and **8**, the modular wall **48** may be used as a portion of a base frame

establishing the habitable first level of the system and spanning the majors subsystem, the minors subsystem, and the discharging subsystem. In the majors subsystem, the system also includes a dispensing level frame constituted from two of the horizontal modular dispensing frames **50** of FIG. **10** situated side-by-side and carried on the base frame, and a storage container frame constituted from eight of the vertical modular container frames **52** of FIG. **11** situated in a 4×2 array carried on the dispensing level frame.

(50) With reference to FIG. **10**, a representative horizontal or dispensing module frame **50** is constructed as a rectangular box truss, having a longitudinal axis L.sub.50, a transverse or lateral axis T.sub.50, and a vertical axis V.sub.50, including lower beams **50***a* extending longitudinally and being laterally opposed from one another, and including upper beams **50***b* extending longitudinally and being laterally opposed from one another. The frame **50** also includes posts **50***c,d* extending vertically between the lower and upper beams **50***a,b* and, more specifically, may include corner posts **50***c* extending vertically between ends of the lower and upper beams **50***a,b*, and intermediate posts **50***d* extending vertically between intermediate portions of the lower and upper beams **50***a,b* between the ends thereof. The frame **50** also includes lower end cross-members **50***e* extending laterally between the lower beams **50***a*, and upper end cross-members **50***f* extending laterally between the upper beams **50***b*. Likewise, the frame **50** also may include lower intermediate cross-members **50***g* extending between portions of the lower beams **50***a* between the ends thereof, and upper intermediate cross-members **50***h* extending between portions of the upper beams **50***b* between the ends thereof. The frame **50** may also include one or more side struts **50***i* extending obliquely between the lower and upper beams **50***a,b* and end struts **50***j* extending between lower and upper end cross-members **50***e,f* opposite longitudinal ends of the frame **50**.

(51) With reference to FIG. **11**, a representative vertical or container modular frame **52** is constructed as a rectangular box truss, having a longitudinal axis L.sub.52, a transverse or lateral axis T.sub.52, and a normal axis N, and including corner beams **52***a* extending longitudinally, and being laterally and normally opposed from one another, and end cross-members **52***b* and intermediate cross-members **52***c* extending laterally and normally between the beams **52***a*. The frame **52** also includes one or more longer struts **52***d* extending obliquely between the beams **52***a* and may be attached to the beams **52***a*. The frame **52** further includes one or more shorter struts **52***e* extending between the beams **52***a* and a corresponding cross-member **52***c*, and one or more intermediate struts **52***f* extending between the beams **52***a* and coupled thereto. Finally, the frame **52** also may include platform brackets **52***g* coupled to upper intermediate cross-members **52***c* and configured to support a platform (not shown) thereon to establish a habitable attic level of the system.

(52) With reference to FIG. **12**, the modular storage container frame **52** can be shipped with or without the associated storage container equipment on a standard seagoing flat rack **57** like a Mafi trailer or the like to constitute a rack and module assembly **58**. On trucks, the modular frame **52** (shipped as part of a module with equipment carried by the modular frame) is designed to be self-supporting and may be wrapped in plastic foil or sheet or truck tarpaulins (not shown) to seal against dust, dirt, and sea water/air, and bottoms and tops may be covered with planks or sheets (not shown) of wood, metal, or plastic to protect the equipment in the module. On ships, the module frame **52** and equipment may be placed on the rack **57** and rolled onto a roll on/roll off ship at a departure seaport and, at an arrival seaport, the rack **57** is rolled off the ship and the module is placed on a truck. Accordingly, the module can be placed in a closed belly of the ship and not be exposed to sea water.

(53) The same can be said for the dispensing modular frame **50** of FIG. **10**, which in some cases can be stacked in pairs one over the other with overall external dimensions equal to or less than those of an intermodal freight container and supported on a standard seagoing flat rack **57** like a Mafi trailer or the like to constitute a rack and module assembly **58**. In fact, the different modular frames **50**, **52** may share one or more common exterior dimensions such as dimensions along their

respective transverse axes T.sub.50, T.sub.52 and be easily aligned with one another to facilitate positioning and assembling them together on site.

(54) With reference again to FIGS. **1**A-**8**, the majors system **38** is configured to receive, pneumatically convey, store, and gravity dispense majors bulk material. Glassmaking majors may include sand, soda, limestone, alumina, saltcake, and, in some cases, dust recovery material. In general, the majors system **38** is the tallest of the subsystems and is supported over a portion of the transport subsystem **44** and thus has its bottom aligned with the bottom of the second level **22** of the system **10** and extends upward to the attic level **24**.

(55) With reference to FIGS. **13-16**, the majors system **38** includes a bulk material storage and dispensing system **100** including an array **200** of bulk material storage container modules **110** atop an array **300** of bulk material dispensing modules **120**. The majors system **38** also includes a pneumatic bulk material receiving and conveyance system **130** including the above-mentioned receiving conduit **39** coupled with a conduit system **132** carried by the storage container array **200**. The storage and dispensing system **100** is both intramodular and intermodular, meaning that each of the different types of modules **110**, **120** are modular amongst their own kind and are additionally modular with one another.

(56) In the illustrated example, each storage container module **110** includes an individual bulk material storage container **112** carried by a corresponding storage container frame **52**, and each dispensing module **120** includes the dispensing module frame **50** with a plurality of dispensing cells **122** defined between dispensing frame crossmembers **50***h*, **50***g*. The dispensing modules **120** are configured to carry a bulk material dispenser **124** in each cell **122**. The intramodularity of the modules **110**, **120** is by virtue of the respective frames **52**, **50** being identical among their own kind. The intermodularity of the modules **110**, **120** is by virtue of certain dimensions of the frames **50**, **52** being the same. In this example, the frame **52** of each storage container module **110** has the same transverse dimension as the frame **50** of each dispensing module **120**, and the longitudinal dimension of each dispensing cell **122** is the same as the width or normal dimension of each storage container module frame **52**.

(57) Accordingly, each dispensing module **120** can support an 1×n array **200** of storage container modules **110**, where n is the number of dispensing cells **122**. Here, each dispensing module **120** includes four dispensing cells **122**, which is the maximum number possible when the module **120** has a longitudinal dimension equal to or less than that of an intermodal freight container and when each container module **110** has a width equal to or less than the height of an intermodal freight container. The same dispenser module **120** can alternatively carry a smaller number of storage container modules **110** with the capacity to add more at a later date. The dispensing module array **300** in this case is a 1×2 array, with each module **120** including four dispensing cells, and the storage container array **200** is a 2×4 array.

(58) FIG. **16** illustrates another modularity feature of the of the modules **110**, **120** in the form of a storage and dispensing module **100**′ which is itself modular. The system **10** of FIGS. **1**A-**8** includes two of the modules **100**′ of FIG. **16** with the capability to add one or more additional dispensing modules **120**, storage modules **110**, or storage and dispensing modules **100**′.

(59) With reference to FIGS. **17-18**, each bulk material storage module **110** may include the container module frame **52**, the bulk material storage container **112** supported within the frame, a platform **114**, a utilities receiver **116**, and a portion of the conduit system **132**. In this example, the bulk material storage container **112** is a silo having a first or upper portion **118**, a second or lower portion **126**, an inlet **128** located along the upper portion for receiving bulk material into the material container, and an outlet **134** located along the lower portion for discharging bulk material from the material container.

(60) The inlet **128** receives bulk material from the conduit system **132**, and each module **110** includes at least a downpipe or vertical inlet conduit section **136** of the conduit system **132** coupled with the inlet **128** and a horizontal connector conduit **138** of the conduit system configured to be

coupled with another portion of the conduit system carried by an adjacent module. Each module **110** includes conduit supports **140** at the top of the frame **52** for supporting the horizontal connector **138** of FIG. **18** as well as other horizontal connectors **138**′ (FIG. **17**) of the conduit system **132** that are merely routed through the module framework to interconnect surrounding modules.

(61) The silo **112** is configured for gravity discharge of the bulk material from the outlet **134**, which is at the bottom of a spout **142** connected to a lower conical part of the lower portion **126** of the silo. The illustrated silo **112** has a shut-off valve **144** in the form of a transverse plate that can be manually or actuator-driven across the outlet **134** to close it for maintenance of the attached dispensing system, for example.

(62) The platform **114** at least partially surrounds the upper portion **118** of the bulk material container **112** and is level with the top of the silo in this example, thus forming a habitable maintenance space between the top of the silo and the top of the frame **52**. The top of the silo **112** includes an access hatch **146**, a filter assembly **148**, a fill-level sensor **150**, a pressure sensor **152**, a high-pressure relief valve **154**, and/or other components. The filter assembly **148** is passive and contains a filter element to remove solids from the air in the silo **112** displaced by incoming bulk material before venting the air to the atmosphere. The filter assembly **148** may double as a vent to permit air exchange between the inside of the container **112** and outside the container during receiving and/or discharging of bulk material. The fill-level sensor **150** may be radar-based and thus detect the real-time amount of bulk material in the silo as well as the instant rate of filling or discharging. Other types of fill-level sensors such as lidar or load cells can be employed. Each of the sensors, gauges, and/or valves of the silo **112** may be in communication with a system controller (e.g., of the controls subsystem **46**) configured to receive information about the storage module **112** and/or to control those connected components in response to the received information or to other received system information.

(63) The utilities receiver **116** in this case is a junction box for connecting electric power to the module to power sensors, gauges, and other equipment and for placing the above-platform components to controllers located elsewhere in the overall system **10**.

(64) As noted above, the storage container module **110** is intramodular, each having the same external dimensions and being configured to be attached side-by-side with up to four other bulk material storage modules and corner-to-corner with up to four other bulk material storage modules. Each module **110** is also sized to fit atop an individual dispensing cell **122** of an underlying dispensing module **120**. When arranged together in the array **200** of the previous figures, the platforms **114** of each module **110** together form a continuous floor or bottom of the maintenance space or attic, where a person is able to access all of the components on top of each silo and the associated pneumatic conduit system **132** all in one space without the need to climb up and down ladders along the side of each individual silo to do so.

(65) FIG. **19** is a top perspective view of the storage container array **200** with portions of the module frames **52** omitted for a full view of the conduit system **132**. The conduit system **132** includes all of the vertical downpipes **136** that lead to each silo inlet **128**, all of the horizontal connector conduit sections **138**, vertical feed pipes or up-pipes **158**A-D, three-way junctions **160** interconnecting some of the horizontal connectors **138**, and valves **162** for regulating the flow of bulk material through the conduit system **132**.

(66) For purposes of illustration of one particular embodiment, the silos **112** of the illustrated array **200** are labelled A-D, indicating four different types of bulk material intended to be received by, stored in, and discharged from each silo **112**. In embodiments in which the system **10** is a glass feedstock handling system, three of the silos (A) may contain sand, two of the silos (B) may contain limestone, two of the silos (C) may contain soda ash, and one of the silos (D) may contain alumina. One vertical feed pipe **158**A-D is dedicated to each different material type, and each of these feed pipes **158**A-D is at an inlet end of the conduit system **132**. Each of the feed pipes **158**A-

D is coupled with a dedicated segment of the pneumatic receiving conduit **39** leading outside the installation **12**, and represents a branch of the conduit system **132**. The feed pipes **158**A-D are located along a side of the array **200** closest to the exterior wall of the installation **12** through which the segments of receiving conduit **39** extend.

(67) Branches leading to a single silo **112**, such as branch **158**D in this case, do not include a three-way junction **160** or valve **162** because the branch exclusively feeds that one silo. Branches leading to more than one silo include a number of junctions **160** equal to one less than the number of silos being fed by that branch and a number of valves **162** equal to (X−1) multiplied by 2, where X is the number of silos being fed by that branch. In this example, the **158**A branch feeds three silos and thus has two junctions **160** and (3−1)×(2)=4 valves **162**. The **158**B-C branches feed two silos each and thus have one junction **160** each and (2−1)×(2)=2 valves each.

(68) With continued reference to FIG. **19** and additional reference to FIG. **20**A, which illustrates the installation **12** of FIG. **1**A with a bulk material transport vehicle **164** delivering bulk material to the majors system, the conduit system **132** including the three-way junctions **160** and valves **162** is operable as part of the receiving and pneumatic conveyance system **130** to provide a bulk material handling method that includes conveying bulk material from a mobile bulk material container **166** into a stationary bulk material container **112** at a glass manufacturing facility. For simplicity in illustration, only a single silo **112** of one bulk material storage module **110** of the array is illustrated schematically inside the installation **12** in FIG. **20**A. As shown in FIG. **20**A, in addition to the conduit system **132** carried by the silo array **200**, the bulk material receiving and pneumatic conveyance system **130** may additionally include one or more pneumatic bulk material inlet conduits **39** extending through a wall of the installation **12**, a plurality of couplings **168** configured to couple a feed conduit **170** of the mobile bulk material container **166** with the conduit system **132** via the inlet conduits **39**, a receiving terminal **172**, valves **174** operable to open and close to connect and disconnect each coupling **168** with the respective inlet conduit **39**, a dense phase pneumatic panel **175**, a controller **176**, indicators **178** to communicate to a user the proper coupling **168** to use, and utility lines **180** coupling the terminal **172** and/or the controller **176** with the valves **174**. In FIG. **20**A, the pneumatic panel **175** and controller **176** are illustrated schematically because they may be located remotely—i.e., somewhere else in the installation **12**.

(69) In this example, the mobile bulk material container **166** is part of a transport truck **164** that is able to pull-up next to the installation without the limitations of rail cars, although rail cars may still be used. The system **130** is designed to pneumatically convey bulk majors materials from the mobile container **166** to one or more silos **112** of the array **200**. Conventionally, it has not been possible to use pneumatic conveying to fill glass majors containers directly because conventional pneumatic conveying is dilute phase conveying in which air pressure at the inlet side of the system blows the bulk material through conduits as fast as the bulk material can be added to the flow of air in the conduits. While this is not problematic with other silo-containing facilities, it is problematic with abrasive glass feedstock materials such as sand and limestone. Conventional dilute phase pneumatic conveying of such abrasive materials quickly wears down the inner wall of the conduit —particularly at 90-degree or other sharp turns of a conduit system.

(70) The pneumatic receiving and conveying system described here uses dense phase pneumatic conveying to address the conduit wear problem. In dense phase conveying, a series of spaced-apart slugs or packets of the bulk material are conveyed through the conduit system **132**. Dense phase conveying operates at a low air velocity in comparison to dilute phase conveying, which keeps the dense slugs of material together while being conveyed. The slower conveyance speed relative to dilute phase conveying significantly reduces conduit wear with abrasive bulk materials. The dense phase system requires an unconventionally high pressure to move the material through the conduit system. Dense phase conveying may for example required inlet pressure on the order of 20-30 psi compared to the relatively low pressure of 10-15 psi required for dilute phase conveying. Dense phase conveying can be somewhat more expensive than dilute phase conveying due to the lower

feed rates and more complex equipment. But in the case of abrasive glass feedstock materials, the additional costs may be at least partly offset by the ability to eliminate subterranean material pits and bulk material elevators from conventional majors feedstock systems. Another benefit of pneumatic conveyance is the ability to operate a closed pneumatic system throughout the installation and, thereby, an essentially dust-free batch house, which is entirely unknown to the glass industry and some other industries that rely on bulk material handling systems.

(71) Mobile bulk material containers such as pneumatic trailers typically used to deliver and unload bulk materials are generally incapable of sustaining the higher hopper and conveyance line pressures required for dense phase conveying, particularly in the United States. While other industries may employ dense phase conveying of particulate materials within manufacturing or processing facilities, pneumatic unloading from a delivery trailer or railcar is typically via dilute phase only. Then, a specialized dense phase system is provided and used only within the manufacturing or processing facility for intra-plant conveyance. Disclosed herein is a pneumatic unloading system in which the bulk material is unloaded directly from a pneumatic trailer or other mobile bulk material storage container and into the silos or other stationary bulk material storage containers via dense phase conveying. Here, the mobile hoppers and conveying lines leading from the mobile delivery vehicle are pressurized at the higher pressure required for dense phase conveying. It has been found during development of this system that fleets of delivery vehicles and pneumatic unloading trailers must be retrofitted, as bulk material delivery companies have balked at requests for high-pressure capable delivery containers.

(72) FIGS. **20**B-**20**D illustrate portions of the pneumatic system **130** of FIG. **20**A in greater detail in order to further explain dense phase conveying and how it is accomplished. FIG. **20**B illustrates the pneumatic inlet conduits **39**, couplings **168**, and valves **174** of the pneumatic system **130** of FIG. **20**A. In addition, FIG. **20**B illustrates high-pressure lines **171** and pulse-pressure lines **173** of the system **130**. FIG. **20**B is also schematically labelled to indicate which branch **158**A-D of the conduit system **132** each inlet conduit **39** may be coupled with where the illustrative array **200** of FIG. **19** is employed. In this example, only two of the four inlet conduits **39** have one of the pulse-pressure lines **173** associated therewith. The number of high-pressure lines **171** is equal to the number of pulse-pressure lines **173**, and a high-pressure coupling **177** is provided at the end of each high-pressure line.

(73) The two branches **158**A, **158**B of the conduit system **132** with associated pulse-pressure lines **173** employ dense phase pneumatic conveying, while the other two branches **158**C, **158**D employ dilute phase conveying. Where the system **10** is a glass feedstock handling system, the most abrasive feedstock materials—e.g., sand and limestone—may be conveyed using the dense phase branches **158**A, **158**B. Each dense phase branch may have a dedicated high-pressure line **171** as illustrated. In this case, one high-pressure line **171** is associated with branch **158**A, while the other high-pressure line **171**′ is associated with **158**B. Likewise, each dense phase branch **158**A, **158**B may have a dedicated pulse-pressure line **173**, **173**′. Each pulse-pressure line is coupled with the respective inlet conduit **39** near the valve **174** that provides or blocks bulk material conveying into the conduit system **132**.

(74) Each high-pressure coupling **177** is adapted to be coupled with the mobile bulk material container **166** such as that of FIG. **20**A via a second conduit between the coupling **177** and the container **166**. The high-pressure line **171** provides the pressure to push the bulk material into the conduit system **132** in dense slugs or segments.

(75) FIG. **20**C illustrates the pneumatic panel **175** of FIG. **20**A in further detail. The panel **175** includes a pneumatic manifold **179** with one air inlet **181** and three air outlets **183**, **185**, **187**. The panel **175** can be located remotely, away from the conduit system **132** and silo array **200**, for example. In one embodiment, the pneumatic panel is located in a module of the minors subsystem **40** (FIG. **1**B), which may be a more centralized location of the installation **12** than the bulk material receiving area. Additionally, the minors subsystem **40** may also employ pneumatic conveying

systems, and locating majors and minors control panel together within the installation can centralize control and maintenance of the two pneumatic conveying systems.
(76) The air inlet **181** is coupled with an air compressor or other pressure source, such as a standard manufacturing plant air pressure system. A first outlet **183** is coupled with one of the high-pressure lines **171** of FIG. **20**B. The associated high-pressure line **171** may be maintained at 20-30 psi from the first outlet **183** and pressurizes the mobile bulk material container when coupled therewith. Various pneumatic components between the inlet **181** and first outlet may include a 2-way magnetic valve **183***a*, pressure regulator **183***b*, check valve **183***c*, pressure sensor/gauge **183***d*, and safety valve **184***e*. The magnetic or solenoid valve **183***a* may be controlled remotely via a controller or control system and is one of the valves that may be opened once the associated high-pressure line **171** is coupled with the delivery vehicle **164** or other mobile bulk material storage container **166**. Other types of remotely controllable valves may be employed, and the panel **175** may include additional components between the inlet **181** and first outlet **183**.
(77) The second outlet **185** of the manifold **179** is coupled with one of the pulse-pressure lines **173** of FIG. **20**B. The associated pulse-pressure line **173** has a non-uniform pressure during conveying with periodic high-pressure pulses and an otherwise low or zero baseline pressure between pulses. Various pneumatic components between the inlet **181** and second outlet **185** may include a pressure regulator **185***a*, a 2-way magnetic valve **185***b*, a volume regulator **185***c*, and a check valve **185***d*. Pressure pulses can be generated via a time-dependent or otherwise controllable valve of the panel **175**, such as the magnetic valve **185***b*, which can be controlled remotely via a controller or control system. The valve **185***b* is closed for a period of time to allow pressure to build and then opens to discharge the built-up pressure before closing again to build pressure for the next pulse. The valve **185***b* can be electronically controlled and electrically operated, or it may be similar to a pressure relief valve that mechanically opens at a threshold pressure and closes again when the pressure drops. There may be other suitable techniques for generating pressure pulses at the second outlet **185** and in the pulse-pressure lines **173**.
(78) The effect of pressure pulses in the pulse pressure lines **173** is the injection of periodic air pockets into the stream of bulk material being conveyed into the conduit system **132** with dense slugs of bulk material between successive air pockets. Those air pockets between slugs of bulk material are compressed during conveyance, effectively keeping the entire length of conduit periodically pressurized rather than just being pressurized at the inlet.
(79) The third outlet **187** is coupled with a dense phase boost line **189** (FIG. **20**D). The boost line is optional but useful in dense phase conveying where a portion of the conveyance is vertical, since some of the conveying energy is lost to potential energy in the higher regions of the system. When employed, such boost lines **189** may be coupled with the conduit system **132** at one or more different heights to inject additional air pockets into the stream of bulk material and/or repressurize already existing air pockets. Various pneumatic components between the inlet **181** and third outlet **187** may include a 2-way magnetic or other remotely controllable valve **187***a*, a pressure regulator **187***a*, and/or other components. A controller or control system can remotely operate the valve **187***a* and/or synchronize its operation with the pressure pulses in the associated pulse-pressure line **173**.
(80) The pneumatic panel **175** may of course include other common pneumatic components such as pressure regulators, shut-off valves, flow restrictors, and/or sensors. Each branch **158**A-D in which dense phase conveying is desired may have a dedicated pneumatic panel.
(81) FIG. **20**D is a schematic representation of a mobile bulk material container **166** coupled with the pneumatic receiving and conveying system **130** to convey bulk material into a bulk material silo or stationary storage container **112** via dense phase conveying. The mobile storage container **166** is coupled with the pneumatic inlet conduit **39** via a bulk material feed conduit **170** and coupling **168**. The mobile storage container **166** is pressurized by the high-pressure line **171**, to which it is coupled via coupling **177**. The high-pressure line **171** is pressurized from the pneumatic panel **175**. The pulse-pressure line **173**, powered by the pneumatic panel **175**, is coupled with the

bulk material inlet conduit **39** downstream of the valve **174**. And a boost pressure line **189** powered by the pneumatic panel is coupled with the conduit system **132** at multiple points along the conduit between the inlet conduit **39** and the storage container **112** into which the bulk material is being conveyed.

(82) A bulk material handling method may include one or more of the following steps in various operable orders. In one aspect, the system **130** employs non-human verification that the bulk material from the mobile storage or transport container **166** is of the type intended to be stored in the stationary container **112** with which the mobile container is coupled. In one example, non-human verification includes receipt at the receiving terminal **172** of information pertinent to the type of material contained in the mobile container **166** before conveying begins. For instance, a 2D data matrix, QR code, bar code, or other encoded machine-readable image can be included on a bill of lading or other shipping document in the transport vehicle **164** operator's possession upon delivery. The terminal **172** may include a camera or other type of scanner configured to recognize the image on the shipping document and match the image with a material type from a data table in computer memory, for example. In other examples, the delivery vehicle **164** may be equipped with an RFID tag or other wireless communicator indicative of the type of bulk material contained in the mobile container, and the receiving terminal is an RFID reader or wireless receiver that does not require human-user interaction.

(83) In one manner of operating the pneumatic receiving and conveyance system, the driver/operator of the transport vehicle **164** arrives at the installation **12** with the mobile container **166** filled with a particular type of bulk material. The driver/operator may be unaware of the type of bulk material being delivered. On arrival, the driver/operator presents a shipping document to a vision system of the receiving terminal **172**, which reads a graphic image and thereby determines the type of material in the mobile container **166** and actuates the appropriate indicator **178** to inform the driver/operator which of the multiple couplings **168** along the outside of the installation **12** is the proper coupling to receive the type of material in the mobile container **166**. In FIG. **20**A, the indicators **178** are visual indicators and the leftmost indicator in the figure is shown illuminated. Auditory indicators can be used in addition to or instead of visual indicators. When the driver/operator couples the feed conduit **170** to the indicated coupling **168**, the system detects the coupling (e.g., via microswitch, capacitive sensing, proximity switch, etc.) and may provide another indicator prompting the driver/operator to approach the receiving terminal, where they are instructed to again present the shipping document to the receiving terminal **172** to verify that the proper coupling **168** has been engaged by the feed conduit **170**. The system **130** may for example employ a mechanical microswitch, a capacitive sensor, a proximity switch, or other type of sensor at each coupling to detect which coupling has been engaged with the feed conduit. If the proper coupling **168** has been engaged, the system **130** checks to determine whether the appropriate high-pressure line **171** has been coupled with the mobile storage container **166**. If not, the driver/operator is provided another indicator to do so. Once the feed conduit **170** is coupled with the proper inlet conduit **39** and with the proper high-pressure line **171**, the corresponding valve **174** is opened and the pneumatic conveying is permitted to begin.

(84) The determinations made by the system **130** based on information received at the receiving terminal **172** can be made locally by a controller at the terminal or remotely by a different system controller, such as a controller **176** of the controls subsystem **46**. Likewise, operation of the valves **174** may be under the control of a terminal controller or another system controller.

(85) In another aspect, the system **130** is capable of selecting which silo **112** of the array **200** the incoming bulk material should be routed to and is further capable of operating the valves **162** of the conduit system **132** to direct the bulk material to the desired silo. In one embodiment, the controller **176** determines which one of multiple silos containing the same bulk material the incoming bulk material should be routed to based on information received from the fill-level sensors. For example, the controller **176** may have information from the various fill-level sensors **150** of the array **200**

allowing it to determine which of the silos **112** containing bulk material A is at the lowest level and which of the same silos is at the highest level. The controller **176** may then control the valves **162** to initially route incoming bulk material to the silo containing the least amount of bulk material A. In other embodiments, the controller **176** may operate to initially top-off the silo **112** having the greatest amount of bulk material contained therein when conveying begins.
(86) In either case, the system **130** may also be configured to reroute incoming bulk material to a different silo containing the same type of bulk material during conveyance without interrupting the conveying. An example is illustrated schematically in FIGS. **21-23**, where three of the silos **112** are intended to contain bulk material A and are labelled A**1**-A**3**. Here, a mobile bulk material container **166** has arrived for delivery of bulk material, and information has been received at the receiving terminal **172** that the type of bulk material is bulk material A. The controller **176** receives that information and, in response, determines which of silos A**1**-A**3** has the least amount of material inside, based on information received from the fill-level sensors on each silo. In this case, silo A**3** contains the least amount of bulk material, and the controller opens the valves **162** leading from branch feed pipe **158**A to the downpipe **136** of silo A**3** while closing the valves leading to the downpipes of silos A**1** and A**2** so that the incoming material is initially routed to silo A**3**, as indicated in FIG. **21**.
(87) Once the fill-level sensor of silo A**3** indicates that a threshold level of bulk material is contained in silo A**3**, the controller reroutes the incoming material to one of the other two silos A**1**-A**2** based on which of the silos presently contains the least amount of material. In the example of FIG. **22**, the incoming material is rerouted to silo A**2**. This involves first opening the valve **162** leading to the downpipe **136** of silo A**2**, and then closing the valve leading to silo A**3** while leaving the other two valves in the same position. This order of valve operation helps prevent unwanted pressure build-up un the conduit system **132**.
(88) Once the fill-level sensor of silo A**2** indicates that a threshold level of bulk material is contained in silo A**2**, the controller reroutes the incoming material to silo A**1** if there is still bulk material remaining to be conveyed from the mobile container **166** and if silo A**1** is not already at the threshold level indicating it is full. In any case, conveying is halted if all silos containing the same type of material are full. In FIG. **23**, the incoming material is rerouted from silo A**2** to silo A**1**, which involves first opening the valve **162** leading to the downpipe **136** of silo A**1**, and then closing the valve leading to silos A**3** and A**3**.
(89) This order of silo filling is merely illustrative, and other valve control schemes can be used. For example, the fill-level of each of a plurality of silos containing the same bulk material type can be monitored during conveying from the mobile bulk material container and the valves of the system **132** can be controlled to more evenly distribute the incoming bulk material.
(90) With reference now to FIGS. **13-16** and FIGS. **24-34**, various components of an illustrative bulk material dispensing module **120** are described in further detail. FIGS. **24** and **25** respectively illustrate top and bottom perspective views of a dispensing module **120** of the dispensing module array **300** of FIGS. **13-15** and of the storage and dispensing module **100′** of FIG. **16**. Each dispensing module **120** includes the dispensing module frame **50** of FIG. **10** and one or more bulk material dispensers **124**. Dispensing cells **122** are defined between successive transverse cross-members **50***h*, which are spaced apart by the width of the silo modules **110** which they support. In this example, a bulk material dispenser **124** is supported by the frame **50** in each of the four dispensing cells **122**. One end of each dispenser **124** is supported in its respective cell **122** by upper and lower intermediate cross-members **50***g,h*, and an opposite end of each dispenser is supported by an additional transverse member **182** having its ends affixed to the upper beams **50***b* of the frame **50**. Each dispensing cell **122** also includes one or more microcontrollers **184** on one vertical side (e.g., the back side) of the frame **50** and a pressure valve **186** on the opposite vertical side (e.g., the front side) of the frame.
(91) With continued reference to FIGS. **24** and **25**, each material dispenser **124** includes an inlet

**188** accessible through a first or top side of the frame **50** and configured to be coupled with and receive material from the bulk material container **112** of an overlying storage container module **110**. An outlet **190** of each dispenser **124** is accessible through an opposite second or bottom side of the frame **50** and configured to be coupled with and discharge bulk material to a transport bin (not shown in FIGS. **24-25**). Each bulk material dispenser **124** also includes a conveyor **192** configured to move bulk material from the inlet **188** to the outlet **190** when the inlet is coupled with the overlying bulk material container. The conveyor **192** in this case is a screw conveyor comprising a screw with one or more screw flights housed in a housing **194** and rotated within the housing by a motor **196** or other actuator under the control of the associated microcontroller(s) **184**.

(92) Each bulk material dispenser **124** includes a dosing assembly **198** that provides the dispenser inlet **188** and includes the conveyor **192**, as well as a docking assembly **202** that provides the dispenser outlet **190**. The docking assembly **202** is arranged beneath the dosing assembly **198** to receive bulk material therefrom. Each bulk material dispenser **124** also includes at least a portion of a filter assembly **204** configured to remove solids from air inside the dispenser during dispenser operation. Each dispensing module **120** may include only a portion of the filter assembly **204** as part of the stand-alone module **120** due to the height of certain components of the filter assembly causing it to extend above the frame **50** when fully assembled. The illustrated dispensing module **120** thus includes only a lower portion **206** of the filter assembly **204** when the module is built remotely to be shipped to the installation site.

(93) With reference now to FIGS. **26-34**, an illustrative bulk material dispenser **124** is described as fully assembled with the overlying bulk material storage container **112** and dockable with an underlying transport bin **208**. The illustrated bulk material dispenser **124** includes the dosing assembly **198** as an upper portion, the docking assembly **202** as a lower portion, and the filter assembly **204** configured to filter solids from air displaced from the transport bin **208** (FIGS. **28-30**) during dispenser operation. The dispenser inlet **188** is carried by an inlet portion **210** of the dosing assembly **198** and is coupled with the outlet of the associated bulk material container **112** to receive bulk material therefrom. The inlet portion **210** feeds the conveyor **192** and, in this example, includes a hopper **212** coupled with the conveyor **192** at a lower end of the hopper and a connector tube **214** coupling the hopper **212** with the outlet of the overlying storage container **112**. The conveyor **192** moves bulk material received from the storage container **112** from the inlet portion **210** to the outlet **190**.

(94) The dispenser outlet **190** is carried by the docking assembly **202** and is coupled with the transport bin **208** during dispenser operation to discharge the bulk material into the transport bin. The dispenser outlet **190** is provided by a lower plate **216** of the docking assembly **202** in this example. The lower plate **216** and outlet **190** are moveable toward and away from the dosing assembly **198** to couple with and decouple from the transport bin **208**. The docking assembly **202** has an inlet **218** coupled with an outlet **220** of the conveyor **192** and includes one or more actuators **222** that move the dispenser outlet **190** with respect to the docking assembly inlet **218**. In this example, the actuators **222** are pneumatic actuators and, more particularly, are lost-motion actuators configured to limit an amount of force applied to the transport bin **208** during docking and dosing. The docking assembly **202** and its operation will be described further below.

(95) With reference to FIGS. **26-28** and **30**, the filter assembly **204** includes a filter inlet **224**, a filter outlet **226**, a turbine **228**, a housing **230** with an internal filter element, an air pressure accumulator tank **232**, and a solids outlet **234**. The filter inlet **224** is in fluidic communication with the dispenser outlet **190** via a conduit **236** extending between a vacuum port **238** and the filter inlet. More particularly, the filter inlet **224** is in fluidic communication with an internal volume **240** of the docking assembly **202** so that, when coupled with the transport bin **208** with the turbine **228** operating, an internal pressure of the internal volume **240** of the docking assembly **202** is less than the surrounding atmospheric pressure. This low-pressure region **240** within the docking assembly **202** ensures that no dust or other solids in the air displaced from the transport bin **208** during bulk

material dispensing escapes from the system.

(96) With reference to FIGS. **26**, **30** and **32**, an adjustable vent **242** is provided, as part of the docking assembly **202** in this case, to permit atmospheric air to enter the internal volume **240** of the docking assembly during turbine **228** operation and prevent the internal pressure from dropping too low and causing the turbine to be overworked. The illustrated vent **242** includes an annular adjuster **244** with apertures **246** formed therethrough. The adjuster **244** is located atop the lower plate **216**, which has corresponding apertures formed therethrough. The adjuster **244** can be rotated about a vertical axis between a fully open position, in which the apertures **246** of the adjuster are aligned with the apertures of the lower plate **216**, and a fully closed position, in which all apertures are closed-off. Adjustment of the vent **242** between these two extremes results in adjustment of the pressure differential between the internal volume **240** and the surrounding atmosphere. In particular, a more open vent **242** results in a higher internal pressure (and a lower pressure differential with the atmosphere), while a more closed vent results in a lower internal pressure (and a higher pressure differential with the atmosphere. This adjustment can be fine-tuned by starting with a fully open vent **242** and gradually closing it off until the pressure is sufficiently low in the internal volume **240** to prevent dust and other solids from escaping during dispensing.

(97) The filter assembly housing **230** and its internal filter element are arranged between the filter inlet **224** and outlet **226**. In this example, the outlet **224** is provided by the turbine assembly **228**, which has an internal impeller operable to force air from the filter inlet **224**, through the filter element, and out of the filter outlet **226**. Dust and other solids filtered from the displaced transport bin air are routed to the conveyor **192** via gravity through the solids outlet **234** of the filter assembly **204**. To accommodate this capture and rerouting of filtered particulates, the accumulator tank **232** is pulsed or discharged after dosing so that the solids fall to the conveyor **192**. The accumulator tank **232** is charged via a system pressure source between pulse cycles. The filter pulse cycle is effected via the pressure valve **186** of the corresponding dispensing cell **122**.

(98) With reference to the schematically depicted conveyor **192** of FIG. **31**, where the conveyor **192** is a screw conveyor, an internal screw **248** may include a first screw flight **250** at one end of the screw that moves the bulk material in a first direction from the inlet **188** toward the conveyor outlet **220**, and a second reverse screw flight **252** at an opposite end of the screw that moves the solids recovered from the filter assembly **204** from the solids outlet **234** in an opposite second direction toward the conveyor outlet **220** while the screw **248** is being turned in only one rotational direction about its axis.

(99) With reference now to FIGS. **32-33** the docking assembly **202** is further described. The illustrated docking assembly **202** includes a receiving portion **254** that includes the docking assembly inlet **218**, a docking portion **256** that includes the dispenser outlet **190**, and one or more actuators **222** that moving the docking portion with respect to the receiving portion. The docking portion comprises the above-described lower plate **216**, which provides the dispenser outlet **190** and mates with the transport bin **208**. The actuators **222** may be lost-motion actuators as noted above to limit the amount of force applied to the transport bin **208** during docking and dosing. Here, the actuators **222** are pneumatic cylinders, but other actuators and actuator mechanisms are contemplated (e.g., solenoid, servo-powered gear train, etc.).

(100) As best illustrated in the schematic depiction of FIGS. **34-35**, the illustrated docking assembly **202** includes a collapsible sleeve **258** extending between the receiving portion **254** and docking portion **256**. The collapsible sleeve **258** delimits the internal volume **240** of the docking assembly **202**. The collapsible sleeve **258** can be a telescopic sleeve with nesting segments, a corrugated polymer sleeve, a fabric sleeve, or similar. The internal volume **240** of the docking assembly **202** thus changes with relative movement of the receiving portion **254** and docking portion **256**.

(101) The receiving portion **254** of the docking assembly **202** includes a coupling sleeve **260** having a first end **262** attached to the dosing assembly **198** and a second end **264** extending into the

internal volume **240** of the docking assembly **202**, as best illustrated in the cross-sectional view of FIG. **33**. The first end **262** of the coupling sleeve **260** provides the docking assembly inlet **218**. The coupling sleeve **260** further includes an inner sleeve **266** and an outer sleeve **268**, both of which extend from the first end **262** and downward into the internal volume **240** of the docking assembly **202**. The vacuum port **238** extends through the outer sleeve and fluidly connects the filter inlet to the internal volume **240** of the docking assembly **202** via an annular gap between the inner and outer sleeves **266**, **268**. The top end of the inner sleeve **266** is funnel-shaped and receives the bulk material from the conveyor **192**. The bulk material thus travels through the center of the docking assembly **202** from the dosing assembly **198** to the transport bin **208**. The inner sleeve **266** extends downward past the end of the outer sleeve and isolates the discharged bulk material from the outer sleeve **268** so that bulk material from the conveyor outlet **220** is not inadvertently drawn into the conduit **236** of the filter assembly.
(102) In addition to the lower plate **216** and adjustable vent **242**, the docking portion **256** of the docking assembly **202** also includes an upwardly extending sleeve **270** to which the lower end of the collapsible sleeve **258** is affixed. All of the sleeves **258**, **266**, **268**, **270** are concentric. When the docking portion **256** is retracted toward the receiving portion **254**, the inner sleeve **266** and outer sleeve **268** of the coupling sleeve are nested within the sleeve **270** of the docking portion **256** and the collapsible sleeve **258** is collapsed. When the docking portion **256** is extended away from the receiving portion **254**, the inner sleeve **266** and outer sleeve **268** of the coupling sleeve are withdrawn from the sleeve **270** of the docking portion **256** and surrounded by the extended collapsible sleeve **258**.
(103) The above-described dispensing equipment enables bulk material dispensing methods, including methods of docking a transport bin with the dispensing equipment and methods of metering doses of bulk material from the bulk material silos at least as follows.
(104) An illustrative bulk material handling method may include a coupling or docking step, a receiving step, formation of a reduced pressure region, and a dispensing step. In the coupling or docking step, the outlet of the bulk material dispenser **124** is coupled with a transport bin **208** to form a closure at an inlet of the transport bin and place an inside of the transport bin in communication with the dispenser. The dispenser **124** and transport bin **208** are illustrated in the docked or coupled condition in FIGS. **28** and **30**, and the inlet of an illustrative transport bin **208** is illustrated in FIG. **29** before docking. In this example, the coupling includes interfacial contact between the lower plate of the docking assembly and a lip surrounding the inlet of the transport bin. Other types of coupling are contemplated, such as positive engagement of protrusions and corresponding recesses, or positive engagement of a latch or other reversible attachment.
(105) The receiving step in this case includes receiving bulk material in the dispenser **124** from the overlying bulk material container **112**. Receiving of the bulk material in the dispenser occurs via gravity feed whenever the conveyer is actively moving bulk material toward the conveyor outlet. Formation of the reduced pressure region occurs in the internal volume **240** of the dispenser **124** when the turbine of the filter assembly is activated. Dispensing of the bulk material occurs via operation of the conveyor, which drops the bulk material from the conveyor outlet, through the reduced pressure region of the internal volume **240**, and into the transport bin.
(106) In one illustrative and more detailed example of the method, the transport bin **208** is placed beneath the docking assembly with the docking assembly in a retracted condition in which the actuators are in a retracted position and the collapsible sleeve is collapsed. With the docking assembly in this state, the dosing assembly and its conveyor are idle and not moving or actively receiving any bulk material, although the conveyor may be entirely full of bulk material from a previous dosing cycle. In addition, the filter assembly and its turbine are idle when the docking assembly is in the retracted condition.
(107) With the inlet of the transport bin aligned beneath the docking portion of the docking assembly, the actuators of the docking assembly are extended and move the docking portion and

the dispenser outlet toward the transport bin as the collapsible sleeve extends. When the docking portion contacts the transport bin and a minimal force is applied, the downward motion of the docking portion is halted by virtue of the lost-motion actuators, and the docked or coupled condition of FIGS. **28** and **30** is achieved.

(108) After the docking assembly and transport bin are coupled together, the turbine of the filter assembly is activated. This reduces the pressure within the internal volume of the docking assembly and, thereby, within the transport bin. With this internal pressure sufficiently reduced, the conveyor of the dosing assembly is activated and begins moving the bulk material received from the overlying silo toward the conveyor outlet, where it is dropped through the concentric sleeves of the docking assembly and into the transport bin. The bulk material discharged from the conveyor is continuously replenished via gravity feed from the overlying silo.

(109) When the desired dose of bulk material is dispensed into the transport bin, the conveyor is deactivated, thereby halting bulk material dispensing. The filter assembly may continue to operate for several seconds after dispensing is halted to remove as much solid material from the air inside the transport bin as possible. The filter assembly is then deactivated, and the filter element may be pulsed to dislodge the filtrate from the filter element to be dropped into the conveyor for dispensing during the next dosing cycle. Next, the actuators of the docking assembly are retracted, and the docking portion of the docking assembly is moved back toward the receiving portion to the retracted position. The transport bin can then be transported to another part of the majors or minors section of the installation.

(110) In various embodiments, the dispensing step includes at least two sequential stages, a later one of the stages being slower than an earlier one of the stages. For example, the conveyor may operate with at least two rotational speeds, including a high speed and a low speed. When the conveyor is initially activated after docking, it may operate at the high speed and then change to the low speed at some threshold amount of the full dose of bulk material. In one particular example, the screw of a screw conveyor rotates at a high rotational speed until 85-95% of the desired dose of bulk material is dispensed, after which the rotational speed of the screw is slowed to a slow speed. The associated “coarse” and “fine” dispensing combines the speed of the high speed dispensing with the accuracy of low speed dispensing, which is most important as the amount of material dispensed into the transport bin approaches the total desired amount.

(111) In various embodiments, the filter assembly may also operate with at least two sequential stages, a later one of the stages being more powerful than an earlier one of the stages. For example, the turbine of the filter assembly may operate with at least two rotational speeds, including a high speed and a low speed. When the turbine is initially activated after docking, it may operate at the low speed to achieved just enough of a reduced pressure region within the docking assembly as is necessary to prevent dust from escaping the coupled system. Then, the turbine may change to the high speed after dosing is completed and the conveyor is deactivated. The high-speed operation draws a much higher volume of atmospheric air through the vent of the docking assembly and causes turbulent flow within the space over the dispensed material in the transport bin to help draw as much of the solids-laden air from the transport bin as possible before halting the vacuum filtration and undocking from the transport bin.

(112) The docking assembly may also cooperate with the transport bin to further reduce the amount of dust and other solids that escape the system during docking and undocking. In one non-limiting example, and with reference to FIG. **29**, the transport bin **208** may be equipped with a closure **272** that is changeable between a closed condition and an open condition. In the example of FIG. **29**, the closure **272** includes a pair of doors, one of which is in the closed condition (i.e., the left door in the figure) and one of which is in the open condition (i.e., the right door in the figure). Levers **274** are affixed to hinges of the doors and extend above the inlet of the bin when the docking assembly is in the retracted condition. The doors of the closure **272** are biased toward the closed condition so that they are closed when the transport bin is undocked. As best shown in the

schematic views of FIGS. **34** and **35**, when the docking assembly is changed from the retracted condition of FIG. **34** to the extended condition of FIG. **35**, the lower plate of the docking assembly contacts the levers **274**, which rotates the doors of the closure to their open condition as the transport bin is docked. Likewise, after bulk material dispensing is completed and the docking assembly is changed back to the retracted condition of FIG. **34**, the doors of the closure are moved back to the closed condition by virtue of their bias toward that condition.
(113) As used in herein, the terminology “for example,” “e.g.,” for instance,” “like,” “such as,” “comprising,” “having,” “including,” and the like, when used with a listing of one or more elements, is to be construed as open-ended, meaning that the listing does not exclude additional elements. Also, as used herein, the term “may” is an expedient merely to indicate optionality, for instance, of a disclosed embodiment, element, feature, or the like, and should not be construed as rendering indefinite any disclosure herein. Moreover, directional words such as front, rear, top, bottom, upper, lower, radial, circumferential, axial, lateral, longitudinal, vertical, horizontal, transverse, and/or the like are employed by way of example and not necessarily limitation.
(114) Finally, the subject matter of this application is presently disclosed in conjunction with several explicit illustrative embodiments and modifications to those embodiments, using various terms. All terms used herein are intended to be merely descriptive, rather than necessarily limiting, and are to be interpreted and construed in accordance with their ordinary and customary meaning in the art, unless used in a context that requires a different interpretation. And for the sake of expedience, each explicit illustrative embodiment and modification is hereby incorporated by reference into one or more of the other explicit illustrative embodiments and modifications. As such, many other embodiments, modifications, and equivalents thereto, either exist now or are yet to be discovered and, thus, it is neither intended nor possible to presently describe all such subject matter, which will readily be suggested to persons of ordinary skill in the art in view of the present disclosure. Rather, the present disclosure is intended to embrace all such embodiments and modifications of the subject matter of this application, and equivalents thereto, as fall within the broad scope of the accompanying claims.

## Claims

1. A bulk material dispenser, comprising: a dispenser inlet configured for coupling with and receiving bulk material from an outlet of a bulk material storage container; a dispenser outlet configured for coupling with and discharging the bulk material into a transport bin; a conveyor that moves bulk material received at the dispenser inlet toward the dispenser outlet; and a filter assembly configured to filter solids from air displaced from the transport bin during dispenser operation, wherein the conveyor receives bulk material via the dispenser inlet at one end of the conveyor and receives the solids filtered from the air at an opposite end of the conveyor via a solids outlet of the filter assembly.
2. The dispenser of claim 1, further comprising: a dosing assembly comprising the dispenser inlet and the conveyor; and a docking assembly comprising the dispenser outlet, wherein the dispenser outlet is moveable toward and away from the dosing assembly to couple with and decouple from the transport bin.
3. The dispenser of claim 2, wherein the docking assembly comprises an inlet coupled with an outlet of the dosing assembly and an actuator that moves the dispenser outlet with respect to the docking assembly inlet.
4. The dispenser of claim 2, wherein an inlet of the filter assembly is coupled with an internal volume of the docking assembly so that an internal pressure of said internal volume is less than atmospheric pressure when the dispenser is coupled with the transport bin.
5. The dispenser of claim 4, further comprising an adjustable vent that permits air flow into said internal volume, wherein adjustment of the vent changes said internal pressure.

6. The dispenser of claim 1, wherein the filter assembly comprises: a filter inlet in fluidic communication with the dispenser outlet; a filter outlet; a filter element between the filter inlet and filter outlet; a turbine arranged to force air from the filter inlet, through the filter element, and toward the filter outlet; and the solids outlet coupled with the conveyor, wherein the solids filtered from the air in the filter assembly are routed to the conveyor for movement toward the dispenser outlet.
7. The dispenser of claim 1, wherein the conveyor is a screw conveyor comprising a screw having a first flight that moves bulk material in a first direction from the dispenser inlet toward the dispenser outlet and second flight that moves filtered solids in an opposite second direction from the solids outlet toward the dispenser outlet.
8. A bulk material dispensing module, comprising: a dispensing module frame having a longitudinal axis, the frame further comprising a plurality of transverse frame members spaced along the longitudinal axis, wherein a dispensing cell is defined between each pair of transverse frame members; at least one bulk material dispenser according to claim 1 supported within the frame, each bulk material dispenser being supported in a different dispensing cell, wherein the dispenser inlet is accessible through a first side of the frame, and the dispenser outlet is accessible through an opposite side of the frame; and a controller carried by the frame for each bulk material dispenser, wherein the module is configured to be attached side-by-side with one or more other bulk material dispensing modules, each of the modules having identical frames, dispenser inlets, and dispenser outlets, and wherein the module has external dimensions less than or equal to an intermodal freight container.
9. A dispensing module array comprising a plurality of the dispensing modules of claim 6 arranged side-by-side with a top and a bottom of each frame lying in common respective planes.
10. A bulk material storage and dispensing system comprising a storage module array arranged atop the dispensing module array of claim 9, wherein each of a plurality of bulk material storage modules of the array is aligned with a different one of the dispensing cells, and each dispenser inlet is coupled with the outlet of a respective bulk material storage container.
11. A glass manufacturing facility comprising a ground level floor and no basement, a main frame on the floor, and the system of claim 10 supported from below by the main frame such that a habitable space is defined between the floor and the system.
12. The dispenser of claim 3, wherein the actuator is a lost-motion actuator that limits an amount force applied to the transport bin by the docking assembly.
13. The dispenser of claim 2, wherein the docking assembly further comprises a collapsible sleeve extending between a receiving portion and a docking portion of the docking assembly and defining an internal volume of the docking assembly that changes with movement of the receiving portion relative to the docking portion.
14. The dispenser of claim 13, the docking assembly further comprising a vacuum port in fluidic communication with the internal volume defined by the collapsible sleeve to reduce an internal pressure of the internal volume when the docking portion is in contact with the transport bin and a negative pressure is applied at the vacuum port.
15. The dispenser of claim 14, further comprising a vent that is operable to adjust the magnitude of the internal pressure for a given applied negative pressure.
16. The dispenser of claim 14, wherein the receiving portion further comprises a coupling sleeve comprising the vacuum port, a first end of the coupling sleeve being configured for attachment to the dosing assembly, and a second end of the coupling sleeve extending into said internal volume.
17. The dispenser of claim 16, wherein the coupling sleeve comprises an inner sleeve and an outer sleeve, the vacuum port being located on the outer sleeve and the inner sleeve being arranged to isolate the bulk material from the outer sleeve.
18. The dispenser of claim 1, wherein the filter assembly comprises a housing and internal filter element located above the conveyor such that the solids filtered from the air can be returned to the

conveyor via the solids outlet of the filter assembly under the force of gravity.
19. The dispenser of claim 1, wherein the conveyor is a screw conveyor comprising a screw that moves bulk material received via the dispenser inlet in a first direction toward the dispenser outlet, and the same screw moves solids filtered from the air in an opposite second direction toward the dispenser outlet.
20. The dispenser of claim 1, further comprising an accumulator tank configured to be pulsed or discharged to cause the solids filtered from the air to fall into the conveyor.
21. The dispenser of claim 1, wherein the filter assembly is operable to force the air from a filter inlet, through a filter element, and out of a filter outlet to atmosphere.
22. The dispenser of claim 1, further comprising an adjustable vent that permits atmospheric air from outside the bulk material dispenser and outside the transport bin to be drawn into the filter assembly along with the air displaced from the transport bin.
23. The dispenser of claim 22, wherein the vent includes at least one aperture and an adjuster movable between a fully closed position, in which each aperture is closed-off, and an open position, in which at least a portion of each aperture is open to permit atmospheric air through the aperture.
24. The dispenser of claim 1, wherein the dispenser outlet is moveable along a straight line toward and away from the conveyor to couple with and decouple from the transport bin.
25. A bulk material dispenser, comprising: a dosing assembly comprising a dispenser inlet configured for coupling with and receiving bulk material from an outlet of a bulk material storage container; a docking assembly comprising a dispenser outlet configured for coupling with and discharging the bulk material into a transport bin, wherein the dispenser outlet is moveable toward and away from the dosing assembly to couple with and decouple from the transport bin; the dosing assembly further comprising a conveyor that moves bulk material received at the dispenser inlet toward the dispenser outlet; and a filter assembly configured to filter solids from air displaced from the transport bin during dispenser operation, wherein the docking assembly comprises an inlet coupled with an outlet of the dosing assembly and an actuator that moves the dispenser outlet with respect to the docking assembly inlet.
26. The dispenser of claim 25, wherein the actuator is a lost-motion actuator that limits an amount force applied to the transport bin by the docking assembly.
27. The dispenser of claim 25, further comprising an adjustable vent that permits atmospheric air from outside the bulk material dispenser and outside the transport bin to be drawn into the filter assembly along with the air displaced from the transport bin.
28. The dispenser of claim 27, wherein the vent includes at least one aperture and an adjuster movable between a fully closed position, in which each aperture is closed-off, and an open position, in which at least a portion of each aperture is open to permit atmospheric air through the aperture.
29. A bulk material dispenser, comprising: a dosing assembly comprising a dispenser inlet configured for coupling with and receiving bulk material from an outlet of a bulk material storage container; a docking assembly comprising a dispenser outlet configured for coupling with and discharging the bulk material into a transport bin, wherein the dispenser outlet is moveable toward and away from the dosing assembly to couple with and decouple from the transport bin; the dosing assembly further comprising a conveyor that moves bulk material received at the dispenser inlet toward the dispenser outlet; and a filter assembly configured to filter solids from air displaced from the transport bin during dispenser operation, wherein an inlet of the filter assembly is coupled with an internal volume of the docking assembly so that an internal pressure of said internal volume is less than atmospheric pressure when the dispenser is coupled with the transport bin, the bulk material dispenser further comprising an adjustable vent that permits air flow into said internal volume, wherein adjustment of the vent changes said internal pressure.
30. A bulk material dispenser, comprising: a dispenser inlet configured for coupling with and

receiving bulk material from an outlet of a bulk material storage container; a dispenser outlet configured for coupling with and discharging the bulk material into a transport bin; a screw conveyor comprising a screw having a first flight that moves bulk material received at the dispenser inlet in a first direction toward the dispenser outlet; and a filter assembly configured to filter solids from air displaced from the transport bin during dispenser operation, wherein the filter assembly comprises a solids outlet coupled with the conveyor, wherein the solids filtered from the air in the filter assembly are routed to the conveyor for movement toward the dispenser outlet, and wherein the screw comprises a second flight that moves filtered solids in a second direction, opposite from the first direction, from the solids outlet toward the dispenser outlet.

31. The dispenser of claim 30, wherein the filter assembly further comprises: a filter inlet in fluidic communication with the dispenser outlet; a filter outlet; a filter element between the filter inlet and filter outlet; and a turbine arranged to force air from the filter inlet, through the filter element, and toward the filter outlet.

32. A bulk material dispenser, comprising: a dosing assembly comprising a dispenser inlet configured for coupling with and receiving bulk material from an outlet of a bulk material storage container; a docking assembly comprising a dispenser outlet configured for coupling with and discharging the bulk material into a transport bin, wherein the dispenser outlet is moveable toward and away from the dosing assembly to couple with and decouple from the transport bin; the dosing assembly further comprising a conveyor that moves bulk material received at the dispenser inlet toward the dispenser outlet; and a filter assembly configured to filter solids from air displaced from the transport bin during dispenser operation, wherein the docking assembly further comprises a collapsible sleeve extending between a receiving portion and a docking portion of the docking assembly and defining an internal volume of the docking assembly that changes with movement of the receiving portion relative to the docking portion, the docking assembly further comprising a vacuum port in fluidic communication with the internal volume defined by the collapsible sleeve to reduce an internal pressure of the internal volume when the docking portion is in contact with the transport bin and a negative pressure is applied at the vacuum port.

33. The dispenser of claim 32, further comprising a vent that is operable to adjust the magnitude of the internal pressure for a given applied negative pressure.

34. The dispenser of claim 22, wherein the receiving portion further comprises a coupling sleeve comprising the vacuum port, a first end of the coupling sleeve being configured for attachment to the dosing assembly, and a second end of the coupling sleeve extending into said internal volume.

35. The dispenser of claim 34, wherein the coupling sleeve comprises an inner sleeve and an outer sleeve, the vacuum port being located on the outer sleeve and the inner sleeve being arranged to isolate the bulk material from the outer sleeve.

36. A bulk material handling method, comprising: coupling an outlet of a bulk material dispenser with a transport bin to form a closure at an inlet of the transport bin and place an inside of the transport bin in communication with the dispenser; receiving bulk material in the dispenser from a bulk material storage container; forming a reduced pressure region in an internal volume of the dispenser; dispensing the bulk material from the dispenser and into the transport bin through the reduced pressure region, wherein the reduced pressure region is provided by a filter assembly such that air displaced from the transport bin during the dispensing is received in the filter assembly; and filtering solids from the air received in the filter assembly.

37. The method of claim 36, wherein the dispensing step includes at least two sequential stages, a later one of the stages being slower than an earlier one of the stages.

38. The method of claim 36, wherein a conveyor moves the bulk material received from the bulk material storage container in a first direction and solids received from the filter assembly in an opposite second direction.